Pizza theorem — quick proof

problem

I found this in Acheson's *The Wonder Book of Geometry*. It is called the "pizza theorem".

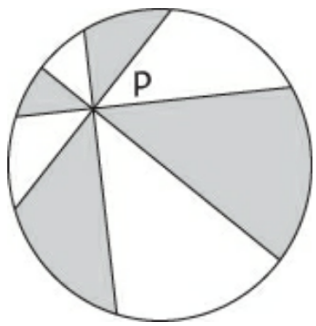


Fig. 111 The pizza theorem.

Consider a circular pizza pie. Choose a point *anywhere* in the disk. Draw two perpendicular chords crossing at the point, with any orientation, and then fill in with two more chords that bisect the angles, giving two pairs of perpendicular chords, rotated at 45°.

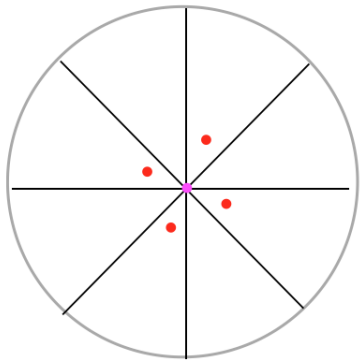
Form the sum of the areas of alternate slices. Above, the two collections are shaded to tell them apart.

The total dark area is always one-half, equal to the total light area. The pizza is evenly divided, even though the slices are wonky.

`https://en.wikipedia.org/wiki/Pizza_theorem`

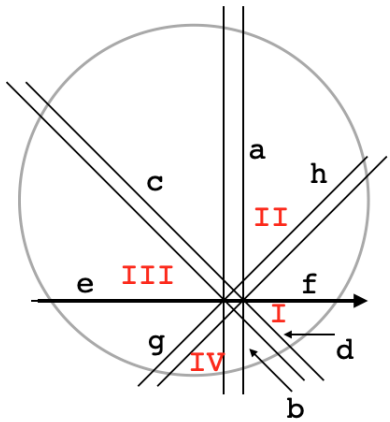
solution

We will refer to the point where all the chords cross as the *grid center*. It is clear that if the grid center coincides with the circle center, then by the radial symmetry of the circle and the equal angles, all the segments are congruent. The equal area result is obviously correct.



For ease of drawing, I've marked the "shaded" slices with dots (and Roman numerals, below).

We will show that any movement parallel to a chord of the grid, whether it is along a diagonal of the circle or not, conserves the equal apportionment of areas between shaded and unshaded regions. The key diagram shows movement to the right in the figure below:



Consider the increase in shaded area due to this movement. That increase comes from rectangles formed by arms of three chords: c, b and h. The decrease in shaded area is due to rectangles on the other end of the same chords: a, d and g. The curved regions at the edges cancel (see below).

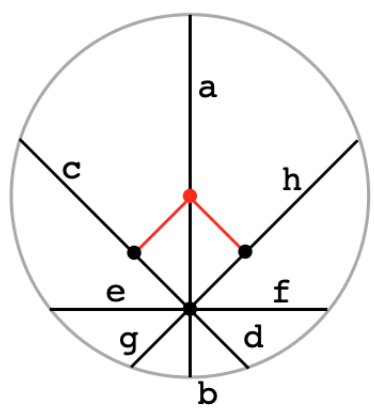
The net change in area is just

$$\Delta A = \sqrt{2}(b - a) + (c - d) + (h - g)$$

The chord ab counts more because it is perpendicular to the direction of travel, so it captures more area.

Our hypothesis is that $\Delta A = 0$ so we should have that

$$\sqrt{2}(a - b) = (c - d) + (h - g)$$



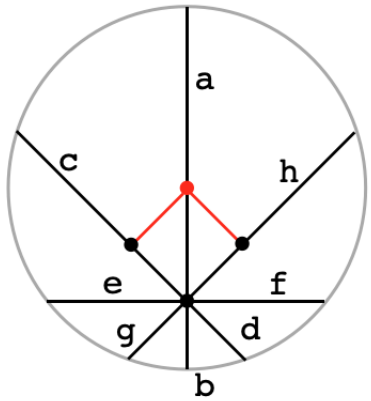
We can find the distance from the center of any chord lies to the grid center by taking the difference between the long and the short arm and dividing by 2.

Call that distance δ. Dividing both sides of what was above by 2 gives the *invariant*: This clearly holds when the grid center remains on a diagonal of the circle (vertical movement). In fact, by symmetry $\delta_{cd} + \delta_{gh}$ so

$$\delta_{ab} = \sqrt{2}\ \delta_{cd}$$

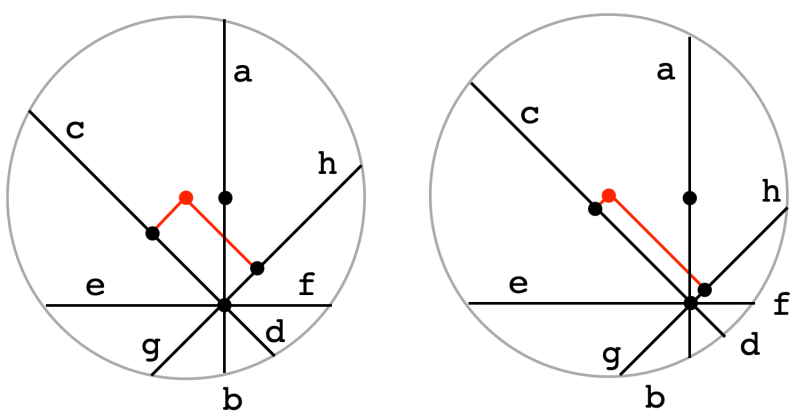
The red lines go from the center of each chord to the center of the circle (red dot). These are perpendicular bisectors of the angled chords. Because of the right angles, these are also equal to δ as defined.

The vertical down from the red dot (center of the circle) to the grid center is δ_{ab}, since the center of this chord coincides with the center of the circle.



It is clear that for this position $\delta_{ab} = \sqrt{2}\ \delta_{cd}$, as we said above. δ_{ab} is the diagonal of a square with sides of length δ_{cd}.

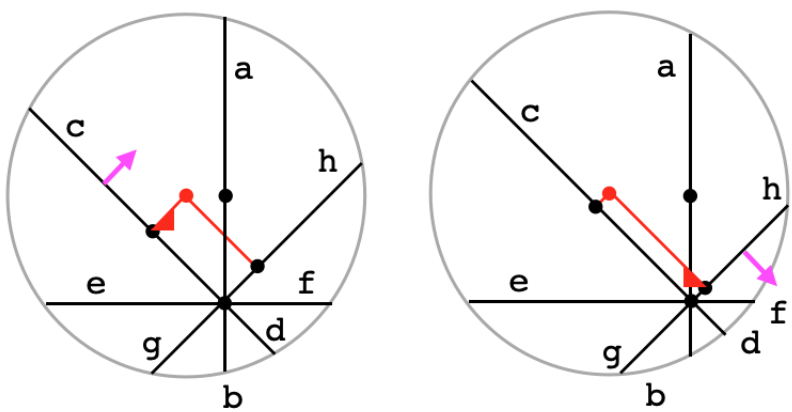
Horizontal movement doesn't change δ_{ab}. Here are two positions reached that way.



It *does* change δ_{cd} and δ_{gh}, but they are adjacent sides in a rectangle

whose four vertices are the chord centers, the circle center and the grid center. It is a rectangle because the two chords are perpendicular and the red lines are perpendicular bisectors.

Put another way, both chords move (maroon arrows, below) along fixed lines connected to the circle center (red lines). The distance each moves for any given horizontal translation is the same (with opposite sign), because the change in the center goes along the hypotenuse of an isosceles right triangle (red triangle), whose base has the magnitude of the movement.



Thus, the total distance from the two chord centers to the center of the circle does not change for horizontal movement. Because of the rectangular arrangement, this distance is equal to $\delta_{cd} + \delta_{gh}$, so the invariant doesn't change either.

summary

We have shown that

$$\sqrt{2} \cdot \delta_{ab} = \delta_{cd} + \delta_{gh}$$

holds regardless of motion horizontally. In fact neither side shows a net change at all. It follows that the areas we have discussed as

shaded and unshaded are also invariant under horizontal translation, even when the grid center is not moving along a diagonal of the circle.

The starting position — grid center at the center of the circle — has the property of equal areas for shaded and unshaded regions, and the transformations do not change the invariant. Therefore, the ending position — grid center anywhere in the circle and with any orientation — also has the property of equal areas.

□

details

◦ Vertical motion along a diagonal doesn't change the allocation of areas, because that diagonal remains an axis of symmetry, where each shaded region has an identical unshaded region across the axis from it.

◦ It is possible to reach any position and orientation by vertical plus horizontal movement. (*Proof*: start with any position and perform a horizontal translation followed by vertical movement along a diagonal of the circle. Now reverse the motion.)

◦ The curved shapes at the ends of each chord, resulting from the motion, are mirror images. Since one changes the shaded and the other the unshaded area, they cancel.

There is an approachable proof for rotation that involves only geometry, here:

`https://math.stackexchange.com/questions/865818/can-anyone-explain-pizz`